MIL-T-18134B(NAVY)
14 August 1980
SUPERSEDING
MIL-T-18134A(NAVY)
11 June 1971
(See 6.4)

MILITARY SPECIFICATION

TORCHES, WELDING, HAND, OXYACETYLENE AND OXYHYDROGEN GASES, LIGHT DUTY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers small sized light duty oxyacetylene and oxyhydrogen hand torches.

1.2 Classification. Torches shall be of the following types as specified (see 6.2):

Type AA - Very light, for oxyhydrogen welding.
Type AB - Light Weight, for oxyacetylene welding.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-6736 - Tubing, Chrome-Molybdenum, 4130 Steel, Seamless, Aircraft Quality.
MIL-Q-9858 - Quality Program Requirements.
MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Alloy Steel Plate Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Design and construction. Torches shall be strongly constructed, easy to repair and assemble, and shall have a minimum number of parts. All parts shall be interchangeable with corresponding parts furnished by the same contractor. A minimum number of repair and adjusting tools shall be required.

3.2.1 Backfire and flash back. Torches shall be free from backfire or flash back that will carry the flame beyond the mixer when the tip comes in contact with the metal or is excessively heated.

3.2.2 Angle of flame. Torches shall be capable of emitting gases at an angle to the axis of the torch body. The angle, unless otherwise specified in the contract or order, shall be between 60 and 70 degrees.

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3.2.3 Tips shall be so threaded that, when changing tips no further adjustment of any other part will be necessary to place the tips in the position providing the greatest possible ease of manipulation in welding.

3.2.4 Mixing. Torches shall mix gases adjacent to or in the forward portion of the handle. The mixer shall be separable from the torch body. Torches with more than one mixer for the range of tips supplied are preferred.

3.2.5 Joints and seats. Joints and seats shall be gas-tight under all working conditions. Joints which are assembled or disassembled in normal operation shall not be dependent upon fit of screw threads for tightness.

3.2.5.1 Joints of the torch body between the rear end and the valve assembly, including the handle, shall be tightly soldered. Soldered joints shall not be affected by the heat of welding.

3.2.6 Balance. When the hose is connected to the inlet nipples of the torch body and a tip is screwed into the torch body, the torch as a whole shall have a good balance in the hand of the welder.

3.2.7 Handle and screws. Torch handles and assembly screws shall be made of corrosion-resistant metal or other nonflammable, corrosion, heat, and impact resistant material.

3.2.8 Tubes. Tubes shall be made of seamless brass or other seamless corrosion-resisting material approved by the command or agency concerned. (Copper tubes for acetylene will not be acceptable.)

3.2.9 Rear end. The torch rear end shall be of wrought brass or other forged corrosion-resisting metal approved by the command or agency concerned.

3.2.10 Valves. Two cut-off valves shall be provided, one for the control of oxygen and one for the control of fuel gas. These valves shall be located at the forward end of the handle and shall be integral with the head of the torch.

3.2.10.1 Valve seats. Valve seats shall be of metal.

3.2.10.2 Handles. Handles shall be made of a corrosion-resisting metal, impact-resistant plastic molded on a metal insert, or other material approved by the command or agency concerned and shall be circular in shape with fluted or knurled gripping surfaces. The valve handles shall not loosen during operation.

3.2.10.3 Valve stems. Valve stems shall be of high-strength bronze, nickel-copper alloy, or Naval brass. The valve-stem assembly shall be provided with retaining washers to prevent the removal of the valve stems from the torch without first removing the packing nuts.

3.2.10.4 Valve packing nuts. Valve packing nuts shall be made of brass or steel. Where the steel selected is not in itself corrosion-resisting, it shall be protected against corrosion by plating.

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3.2.11 Threads. Threads shall be in accordance with FED-STD-H28.

3.2.12 Hose connection nuts. Hose connection nuts shall be made of brass or other corrosion-resisting metal. Threads shall be 3/8-24, NF-3, right hand for oxygen and left hand for fuel gas connection.

3.3 Type AA torches. Type AA torches shall be for use with oxygen and hydrogen gases and for welding aluminum, brass and light sheet steel.

3.4 Type AB torches. Type AB torches shall be for use with oxygen and acetylene gases and for welding steel, aluminum, brass, and other nonferrous metals.

3.5 Length, weight, and range of thickness. The length of torches shall be measured from the outer or flame end of the medium size tip to the forward end of the hose glands. The weight shall include the torch hose connections and the medium size tip. The length, weight and range of thickness shall be as shown in table I.

TABLE I. Length, weight and range of thickness.

Type	Minimum range of metal thickness	Length		Weight	
		Minimum	Maximum	Minimum	Maximum
	Inch	Inches	Inches	Ounces	Ounces
AA	Up to and including 1/8 (aluminum)	10	11	5	6
AB	Up to and including 1/4 (steel)	11	12	8	10.5

3.6 Mixing chamber. Mixing chamber of type AB torch shall be so designed that the torch will consume approximately equal volumes of oxygen and acetylene under pressures recommended by the manufacturer.

3.7 Flame. Unless otherwise specified in the contract or order, the flame produced by the type AB torch shall vary in size according to the size of tip in use, but all flames shall be long and pointed.

3.8 Hose connections. Hose connections shall be securely fastened to the rear end of the torch. Nondetachable hose glands shall be permissible, but threaded connections shall be preferred. The hose connections for oxygen shall be threaded 3/8-24, NF-3. The connections for acetylene shall be threaded 3/8-24, NF-3, left-hand.

3.9 Type AA torch glands. Type AA torch glands shall be of 1/8 inch inside diameter hose.

3.10 Type AB torch glands. Type AB torch glands shall be 3/16 inch inside diameter hose.

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3.11 Tips. One-piece swaged copper tips are preferred. Tips of more than one piece with a permanent tight joint are acceptable. Tips of the latter type may be of copper or copper and brass, provided no brass extends within 2 inches of the base of the flame. Jointed tips shall have no threaded connection or enlargement of cross-section at the joint between metals. Tips shall measure not more than 2-3/8 inches from the point of emission to the bend. Tips shall be manufactured in a range of orifice sizes which shall encompass the applicable thickness range in table I. A full set of tips, consisting of one of each orifice size required to encompass table I thickness range, shall be provided with each torch unless otherwise specified (see 6.2).

3.11.1 Tip tests. Tips shall operate with the torch for which intended and shall be of such materials, design, and workmanship that no gas leakage or damage due to melting will occur under test conditions specified in 4.4.3.

3.11.2 Replacement tips. Replacement tips shall conform to the requirements specified in 3.12 and 3.12.1.

3.12 Marking. Each torch shall be plainly and permanently marked with the manufacturer's name or trademark and type of torch. Each tip shall be marked in accordance with the manufacturer's standard practice.

3.12.1 Valve marking. The name of the gas which each valve controls shall be marked in a permanent manner on the valve body, valve handle, or torch handle adjacent to the valve.

3.13 Wrench. Each torch shall be supplied with one nonadjustable, steel wrench, for operation of union nuts, removal of parts, and adjustment. The steel shall be corrosion-resisting or protected against corrosion by the use of plating. The head shall be machine finished to size.

3.14 Instructions. An instruction sheet showing pressures recommended for each tip shall be furnished with each torch.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program. The contractor shall provide and maintain a quality program acceptable to the Government for the supplies covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

* 4.2 Quality conformance inspection. Unless otherwise specified (see 6.1), quality conformance inspection shall consist of the examinations of 4.3 and the tests of 4.4.

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4.3 Examinations.

4.3.1 In-process examination. Intermediate assemblies which cannot be checked or measured after final assembly shall be examined at various stages of manufacture for compliance with this specification. Whenever a deviation is noted, correction shall be made. Failure to make immediate correction may cause rejection of affected end items.

4.3.2 Visual and dimensional examination. Completely assembled end items shall be examined for defects in material, finish, construction, workmanship and dimensions to determine compliance with specification requirements. When sampling inspection is performed, the inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5 defects per hundred units in accordance with MIL-STD-105.

4.4 Tests.

4.4.1 Torches shall be tested to determine conformance to the following:

(a) The tendency of the torch to backfire and heat excessively when used on flat surfaces and in shielded positions.
(b) The ability of the torch to hold flame settings when used in normal production welding.
(c) The range of flame adjustment with each tip.
(d) The flame characteristics with the various tips.
(e) The ease of assembly and disassembly for cleaning and repair.
(f) The range of sizes of materials which can be welded with the equipment.
(g) The ability of the equipment to weld various aircraft materials.
(h) Speed of welding.

At least three experienced welders shall use the torches in simulating production work in order to determine their performance.

4.4.2 Type AB welding tests. A three-side pyramid-shaped part, with sides of 3/16-inch thick steel sheet in the form of equilateral triangles 3 inches along the edge, shall be fabricated by inside welds. The welds shall be made as quickly as practicable and consistent with good welding practice. The tip shall be removed from the work only as necessary to turn the specimen (not over three times) and shall not be cooled during the entire welding operation. The steel sheet used shall be in accordance with AISI 4130.

4.4.2.1 Tube and plate test. A 2-inch diameter steel tube, 0.063 inch thick wall and 2 inches long, shall be welded inside onto a steel plate 0.062 inch thick, without any backfiring whatsoever. The weld shall be made while the specimen is placed so that the axis of the tube is at a 45-degree angle from the vertical, open end up. The tip shall be removed from the tube only as necessary to turn the specimen (not over two times). No cooling of the tip shall be permitted. The speed shall conform to good welding practices. The steel plate shall conform to AISI 4130 and the tube to MIL-T-6736

4.4.3 Type AA welding tests. A three-side pyramid-shaped part, with sides of 14 to 16 gage (0.0747 to 0.060 inch) steel sheet in the form of equilateral triangles 3 inches along the edge, shall be fabricated by inside

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welds. The welds shall be made as quickly as practicable and consistent with good welding practice. The tip shall be removed from the work only as necessary to turn the specimen (not over three times) and shall not be cooled during the entire welding operation. The steel sheet used shall conform to AISI 4130.

4.4.3.1 Tube and plate test. The tube and plate test shall be as specified in 4.4.

4.4.4 Examination and test of tips. To determine compliance with 3.12 the tip shall be tested to determine whether it operates with the torch for which intended and also whether the tips are of such material, design, and workmanship as to comply with the gas leakage and melting requirements of 3.11. The test for gas leakage shall be made by immersing the torch head in water while the torch tip is lighted and the flame end of the tip held above the water level. The appearance of bubbles shall be taken to indicate gas leakage. The test for welding tip melting shall be made by lowering and holding the lighted torch and tip, operating under optimum pressures of oxygen and acetylene, in a cast iron target having a 1-inch diameter hole approximately 3/4 inch in depth. This test shall continue until either the gases pre-ignite with resultant "popping" or until the tip end melts, after which it shall be determined whether the tips so tested are usable. In cases where neither "popping" nor "melting" is observed after 1/2 hour, this test shall be discontinued. Figure 1 shows an effective arrangement for conducting the tip melting test.

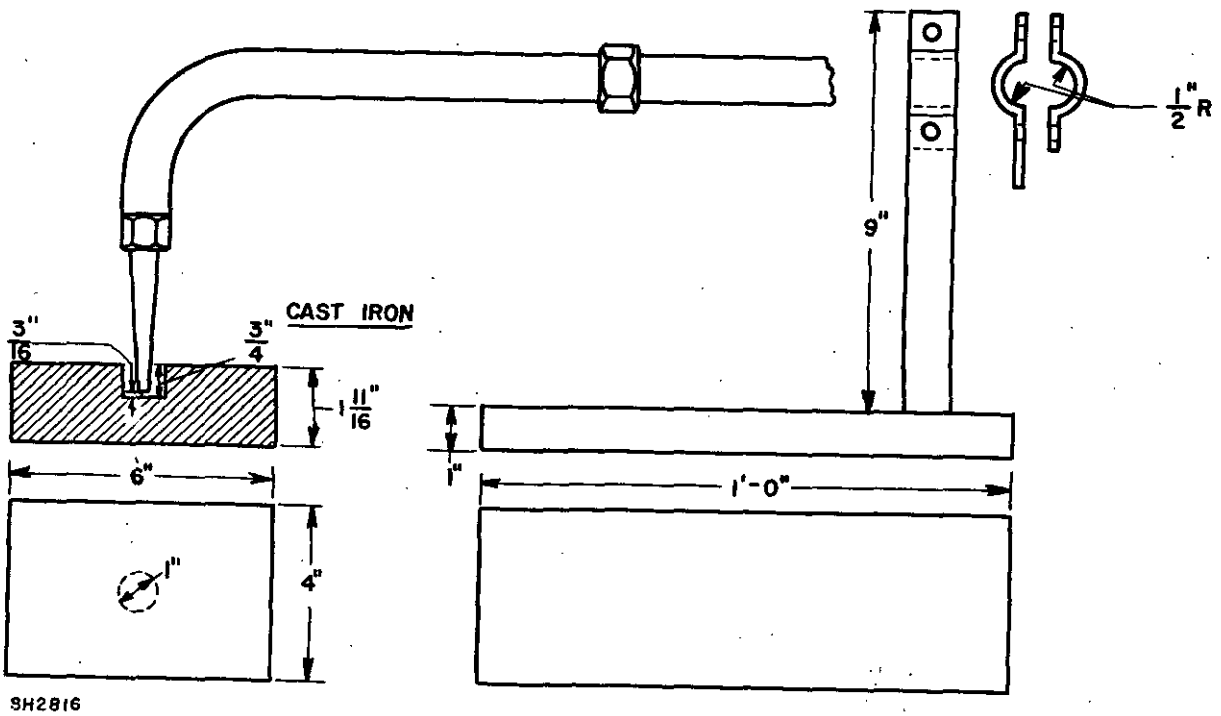


FIGURE 1. Welding tips test equipment.

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4.5 Inspection of packaging. An examination shall be made to determine compliance with the requirements for packaging, packing, and marking, as specified in section 5. When examination is performed, the inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

5. PACKAGING

* 5.1 Preservation-packaging, packing and marking. Each complete torch unit with wrench shall be packaged level A or commercial, packed level A, B or commercial as specified (see 6.2) and marked in accordance with MIL-W-45562, except that in lieu of the method cited in table I of MIL-W-45562, method III preservation-packaging shall apply.

* 5.2 Cushioning, filler, dunnage and wrapping materials.

* 5.2.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler or dunnage is prohibited for materials destined for shipboard installation/stowage.

* 5.2.2 Level commercial preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler and dunnage, all containers (unit, intermediate and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned, etc., with loose-fill material. Not to be taken aboard ship. Remove and discard loose-fill material. If required, recushion with cellulosic material, bound fiber, fiberboard or transparent flexible cellular material."

* 5.2.3 Cushioning, filler, dunnage and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire.

6. NOTES

* 6.1 Intended use. Oxyacetylene and oxyhydrogen torches covered in this specification are to be used for welding, brazing and heating thin sections in confined areas.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Whether one tip of each orifice size is required (see 3.12).
(d) If quality conformance inspection is required (see 4.2).
(e) Level of preservation-packaging and packing required (see 5.1).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.4 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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